Caitlin Coffey

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Education

Bachelor of Science in Engineering with Computing, May 2023, GPA: 3.97/4.00
Franklin W. Olin College of Engineering, Needham, MA

- Recipient of a 4-year, 50% Olin Merit Scholarship
- **Relevant coursework:** Software Systems, Computer Architecture, Data Structures and Algorithms, Machine Learning, Principles of Integrated Engineering
- **Clubs:** Public Interest Technology, Olin Conductorless Orchestra, Society of Women Engineers

Work Experience

The MITRE Corporation, Bedford, MA
Data Science & Network Security Intern 05/2021-08/2021

- Design, analyze, and publish an automated data pipeline in Apache Airflow and PySpark.
- Conduct research on recurrent neural network designs and debug extensive Python code in TensorFlow, Apache Airflow, and Apache Spark.

Qualcomm Inc., Boxborough, MA
Software Engineering Intern 05/2020-08/2020

- Create a software tool to analyze GPU performance in Python using Matplotlib, NumPy, Tkinter, and Pandas.
- Troubleshoot GPU performance using Verilog, Linux, and Vim.

Olin College of Engineering, Needham, MA
Software Design & Modeling and Simulation Teaching Assistant 09/2020-12/2021

- Assist in teaching data analysis/data science concepts. Analyze and debug code in Python and MATLAB.

Volunteer Experience

Code for Boston, Cambridge, MA
Software Developer 12/2019-04/2020

- Build and test a software tool in YAML, HTML, CSS, and Docassemble to help people from low-income backgrounds expunge or seal their criminal records.

Relevant Skills

Python – Object-oriented programming, artificial intelligence (computer vision, audio processing, natural language processing). Extensive experience with Pandas, PySpark, TensorFlow, PyTorch, and Multiprocessing libraries.

C/C++/x86 Assembly - Reverse engineering (Ghidra), object-oriented programming, embedded systems programming (Arduino, Raspberry Pi), virtual machine design (LC-3)

MATLAB – Data analysis, artificial intelligence (facial recognition using K-nearest neighbor, principal component analysis, and eigenvalue decomposition algorithms), audio processing (using Discrete Fourier transforms)

Linux/Docker – Running and managing containers, writing shell scripts, version-control navigation (Git/Perforce)

Hardware Troubleshooting – Oscilloscopes, Multimeters, writing test scripts in Python